

■ Fun Revision Notes: SQL Subqueries!

■ What Are Subqueries?

A subquery is a query inside another query! ■

It helps you break down complex SQL problems into smaller, easier parts.

■ Why Do We Use Subqueries?

■ To get results based on another query's output.

■ To calculate or filter data dynamically.

■ To simplify complex joins and logic.

■ Subquery in WHERE Clause

Use subqueries in WHERE to filter data using another query's result.

Example ■

```
SELECT name FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

■ Finds employees earning more than the average salary.

■ Subquery in Column Expression

Use a subquery to create a new calculated column!

Example ■

```
SELECT name,
(SELECT department_name FROM departments d
WHERE d.dep_id = e.dep_id) AS Department
FROM employees e;
```

■ Subquery as Table Expression

Subqueries can act like temporary tables!

Example ■

```
SELECT department_id, AVG(salary) AS AvgSalary
FROM (SELECT * FROM employees WHERE salary > 40000) AS HighEarners
GROUP BY department_id;
```

■ *Tip: Subqueries make SQL smarter — use them to impress your teacher!* ■